Confidence Without Competence: Calibration Risks in Real-World LLM Deployment Pipelines

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Abstract

The deployment of large language models (LLMs) in production systems increasingly relies on model compression techniques to meet resource constraints. Two dominant pipelines have emerged: Post-Training Quantization (PTQ), which applies quantization to a pre-trained, instruction-tuned model, and Quantization-Aware Fine-Tuning (QLoRA), which fine-tunes within the quantized weight space. While prior work compares these pipelines on task accuracy, their effect on confidence calibration—the alignment between a model's stated confidence and its actual accuracy—remains understudied. Poor calibration poses a direct safety risk in production agents: a model that is confidently wrong is more dangerous than one that is correctly uncertain. In this work, we empirically compare the calibration behavior of both deployment pipelines using Gemma-2-2B as the base model. We evaluate across three standard benchmarks (MMLU, ARC-Challenge, TruthfulQA) using Expected Calibration Error (ECE), Brier Score, Negative Log-Likelihood (NLL), and behavioral metrics including overconfidence rate. We find that the commonly deployed PTQ pipeline configuration exhibits an overconfidence rate of 29.7% and ECE of 0.293, compared to 0.9% and 0.038 for QLoRA—a 7.7x calibration gap despite the PTQ pipeline achieving 23% higher task accuracy. However, this comparison is confounded by differences in instruction tuning scale. We argue that accuracy alone is an insufficient criterion for deployment decisions when calibration safety is at stake.

1. Introduction

The deployment of language models at scale has accelerated the adoption of quantization as a standard compression technique. Four-bit quantized models have become the de facto standard for local and edge deployment, enabling inference on consumer hardware at the cost of some precision. Two deployment paradigms dominate: in Post-Training Quantization (PTQ), a professionally instruction-tuned model is quantized after training; in QLoRA-based deployment, a base model is quantized first and then fine-tuned through low-rank adapters on task-specific data.

Practitioners currently choose between these pipelines based primarily on two criteria: task accuracy and inference speed. A critical dimension is systematically overlooked: confidence calibration. A model is well-calibrated if its stated confidence on a prediction correlates with the probability that the prediction is correct. A model with 80% confidence on a set of questions should answer approximately 80% of them correctly. Miscalibration—particularly overconfidence—is a known failure mode in deployed AI systems, leading to hallucinated facts stated with high certainty, inappropriate downstream decisions by systems relying on model confidence scores, and erosion of user trust.

This paper presents an empirical study comparing the calibration behavior of PTQ and QLoRA deployment pipelines on standard NLP benchmarks. We make the following contributions:

- We evaluate calibration across two real-world deployment pipelines using a controlled base model (Gemma-2-2B), measuring ECE, Brier Score, NLL, and overconfidence rate.
- We find a consistent 7.7x ECE gap and 33x overconfidence gap in favor of QLoRA across all three benchmarks, despite the PTQ pipeline achieving higher task accuracy.
- We discuss the mechanistic implications and identify the confound between pipeline choice and instruction tuning scale as a direction for future controlled ablation.

Our findings suggest that the PTQ deployment pipeline—as currently practiced—is associated with systematically higher overconfidence in our evaluation setting, which may represent a safety risk for production agents, particularly in domains requiring reliable uncertainty quantification.

To support reproducibility, we release all code, configurations, and evaluation scripts. Code is available at: https://github.com/bremsstrahlung-57/ptq-vs-qlora-calibration

2. Related Work

2.1 Confidence Calibration in Neural Networks

Guo et al. (2017) showed that modern neural networks are often poorly calibrated, tending to produce overconfident predictions despite high accuracy. They popularized the Expected Calibration Error (ECE) as a metric for measuring this mismatch and demonstrated that temperature scaling, a simple post-hoc adjustment, can significantly improve calibration. A key takeaway from their work is that predictive accuracy and calibration are not inherently correlated, motivating evaluation settings where models with differing accuracies are compared in terms of confidence reliability.

Subsequent work suggests that instruction fine-tuning can negatively impact calibration. Models trained on instruction-following or human feedback data often adopt more assertive response patterns, increasing confidence even when outputs are incorrect.

2.2 Quantization and QLoRA

Dettmers et al. (2022) introduced 4-bit NormalFloat (NF4) quantization, a format designed to match the approximately Gaussian distribution of pretrained model weights, enabling highly efficient compression with minimal performance degradation. In subsequent work on QLoRA, they showed that models quantized to 4-bit precision can be fine-tuned using low-rank adapters to achieve performance comparable to full-precision fine-tuning, while substantially reducing memory requirements(Dettmers et al., 2024). Their evaluation primarily focused on accuracy, without examining calibration properties.

Post-training quantization (PTQ) methods for large language models have also been explored extensively. Techniques such as GPTQ (Elias Frantar et al., 2022) and AWQ (Ji Lin et al., 2023) improve quantization fidelity and efficiency, typically optimizing for accuracy and inference speed. However, the impact of these methods on model calibration remains largely unstudied.

2.3 Calibration and Safety

Calibration has important implications for AI safety, particularly in domains where model confidence informs downstream decisions. Overconfident incorrect predictions can be especially problematic in settings such as medical decision-making, autonomous systems, and agent-based workflows.

The TruthfulQA benchmark (Lin et al., 2021) was designed to evaluate whether language models produce answers that are not only plausible but factually correct, particularly in the presence of common misconceptions. While not a direct calibration metric, it serves as a useful proxy for assessing the tendency of models to confidently generate incorrect information, making it relevant to calibration-focused analyses.

3. Methodology

3.1 Pipeline Definitions

We compare two deployment pipelines that reflect current real-world practice:

Pipeline A (PTQ): We load *google/gemma-2-2b-it*—Google's instruction-tuned variant of Gemma-2-2B—with 4-bit NF4 quantization applied at inference time via bitsandbytes. This model was trained in full precision on Google's proprietary instruction-following dataset and subsequently quantized. This faithfully represents the PTQ deployment paradigm: fine-tune in high precision, then compress for deployment.

Pipeline B (QLoRA): We load *google/gemma-2-2b* (base model) with 4-bit NF4 quantization, apply LoRA adapters (rank r=16, alpha=32) to all attention projection matrices (q_proj, k_proj, v_proj, o_proj), and fine-tune on 800 samples from the Alpaca instruction-following dataset for 2 epochs using the paged AdamW 8-bit optimizer. This represents the QLoRA paradigm: quantize first, then fine-tune through the quantized weight space.

Property	Pipeline A (PTQ)	Pipeline B (QLoRA)
Base Model	google/gemma-2-2b-it	google/gemma-2-2b
Instruction Tuning	Google (full precision, large-scale)	Custom (QLoRA, 800 Alpaca samples)
Quantization Timing	Post instruction tuning	Prior to instruction tuning
Quantization Format	NF4 4-bit (bitsandbytes)	NF4 4-bit (bitsandbytes)
Adapter	None	LoRA r=16, alpha=32
Training Data	Proprietary (Google)	tatsu-lab/alpaca (800 samples)

Table 1: Pipeline comparison. Both pipelines use identical base model lineage (Gemma-2-2B) and identical quantization format.

We note that pipeline choice is inherently coupled with instruction tuning scale in this setup: Pipeline A uses a large-scale proprietary instruction-tuned model, while Pipeline B is fine-tuned on 800 samples. As a result, observed differences may reflect both quantization order and training regime.

3.2 Evaluation Datasets

We evaluate on three standard benchmarks formulated as 4-way multiple-choice tasks:

MMLU (Hendrycks et al., 2020): 300 questions sampled uniformly across subjects from the test split, covering STEM, humanities, and professional domains. Measures broad knowledge recall.

ARC-Challenge (Clark et al., 2018): 300 questions from the ARC Challenge split, designed to require multi-step reasoning beyond simple retrieval. Harder than MMLU on average.

TruthfulQA (Lin et al., 2021): 300 questions from the mc1 multiple-choice split, specifically designed to elicit false but plausible-sounding answers. Directly measures calibration under adversarial conditions.

All datasets are sampled with seed=42 for reproducibility. The same 300 samples per dataset are used for both pipelines.

3.3 Confidence Extraction

For both pipelines, we extract confidence scores using a unified custom inference loop rather than third-party evaluation harnesses. This ensures methodological consistency across pipelines.

For each question, we construct a prompt presenting the question and four choices labeled A through D. We feed the prompt to the model and extract the logits at the final input token position. We then slice the logits for only the A, B, C, D answer tokens—identified by querying the tokenizer vocabulary directly—and apply softmax over these four values to produce a normalized probability distribution over the answer choices. The model's predicted answer is the argmax; its stated confidence is the corresponding probability.

This approach constrains the model to one of four valid answers and produces calibrated probability estimates relative to the restricted answer space, which is standard practice for multiple-choice calibration evaluation.

One methodological note: Pipeline A uses Gemma's native chat template (apply_chat_template) for prompt formatting, as gemma-2-2b-it was trained with structured turn markers. Pipeline B uses the Alpaca instruction format (### Instruction / ### Response), consistent with its fine-tuning data. Using each model's native format represents how these pipelines would be deployed in practice, and we view this as appropriate. We acknowledge this as a limitation in Section 6.

3.4 Calibration Metrics

We compute the following metrics:

Expected Calibration Error (ECE): Partitions predictions into 15 confidence bins and measures the weighted mean absolute difference between average confidence and accuracy per bin. Lower is better; 0 indicates perfect calibration.

Brier Score: Mean squared error between stated confidence and binary correctness indicator. A proper scoring rule that penalizes both overconfidence and underconfidence. Lower is better.

Negative Log-Likelihood (NLL): Mean negative log probability assigned to the correct answer. Captures the full distributional behavior rather than just the top prediction. Lower is better.

Overconfidence Rate: Proportion of incorrect predictions where stated confidence exceeds 0.7. This is our primary safety-relevant behavioral metric—it measures how often the model is confidently wrong.

Underconfidence Rate: Proportion of correct predictions where stated confidence is below 0.5. Measures excessive hedging.

Confidence Gap: Difference between mean confidence on correct vs. incorrect predictions. A higher gap indicates better self-awareness.

Importantly, restricting the probability mass to the answer choices avoids generation-induced calibration artifacts and yields more stable calibration estimates.

4. Results

4.1 Overall Calibration

Table 2 and Figure 1 present aggregate metrics across all 900 evaluation samples (300 per dataset × 3 datasets) for both pipelines.

Metric	Pipeline B (QLoRA)	Pipeline A (PTQ)	Ratio (A/B)
Accuracy	0.381	0.612	1.61x (A higher)
ECE ↓	0.038	0.293	7.7x worse for A
Brier Score ↓	0.223	0.299	1.34x worse for A
NLL ↓	1.359	1.726	1.27x worse for A
Mean Confidence	0.412	0.904	2.2x higher for A
Conf. when Correct	0.449	0.943	—
Conf. when Incorrect	0.390	0.843	—
Overconfidence Rate ↓	0.009	0.297	33x worse for A
Underconfidence Rate ↓	0.280	0.013	21x worse for B

Table 2: Aggregate calibration metrics across MMLU, ARC-Challenge, and TruthfulQA. ↓ indicates lower is better.

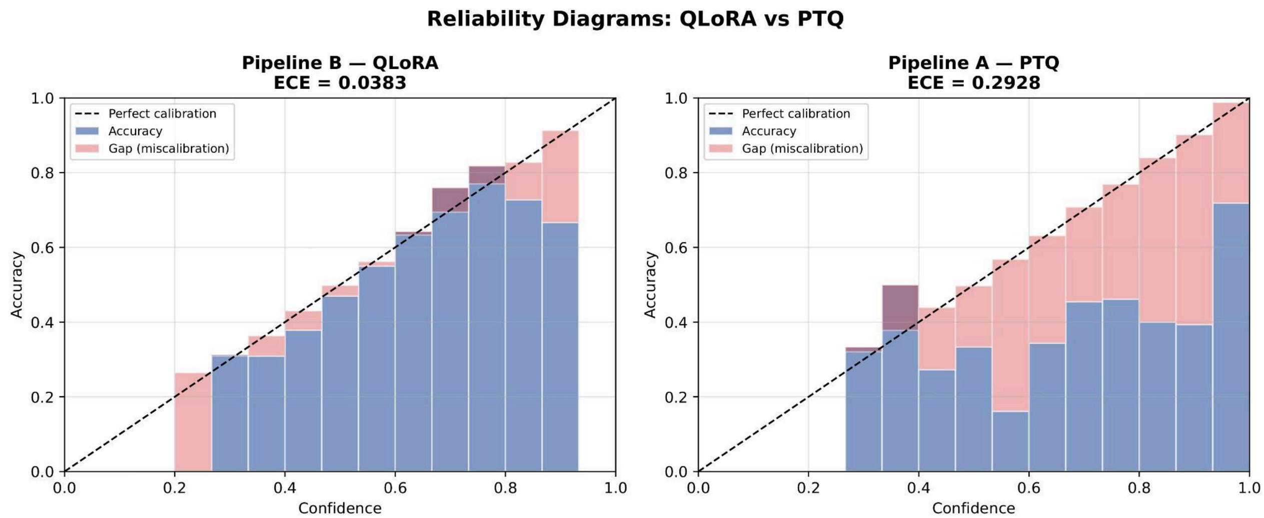

Figure 1: Reliability Diagrams for Pipeline B (QLoRA) and Pipeline A (PTQ)

Pipeline A achieves substantially higher task accuracy (61.2% vs 38.1%), reflecting the difference in instruction tuning quality between a professionally trained model and one fine-tuned on 800 samples. However, across every calibration metric, Pipeline B outperforms Pipeline A substantially.

The most striking finding is the overconfidence rate: Pipeline A is confidently wrong (confidence > 0.7 on an incorrect prediction) on 29.7% of all samples. Pipeline B exhibits this behavior on only 0.9% of samples—a 33x difference. In absolute terms, this means that for every 100 questions Pipeline A answers incorrectly, it expresses high confidence on approximately 30 of them.

The mean confidence gap (0.412 vs 0.904) reveals distinct behavioral profiles. Pipeline B distributes its confidence across the full range, hedging appropriately when uncertain. Pipeline A is nearly always confident—its average confidence is 0.843 even on incorrect predictions, with only a small gap relative to correct predictions (a gap of only 0.10). This suggests that the model has limited ability to distinguish between correct and incorrect predictions.

4.2 Per-Dataset Results

Table 3 and Figure 2 present per-dataset breakdowns. The calibration gap is consistent across all three benchmarks, ruling out dataset-specific artifacts.

Dataset	Pipeline B ECE	Pipeline A ECE	Pipeline B Overconf.	Pipeline A Overconf.	Pipeline B Acc.	Pipeline A Acc.
MMLU	0.05	0.36	0.01	0.34	0.37	0.51
ARC-Challenge	0.09	0.21	0.00	0.21	0.51	0.75
TruthfulQA	0.16	0.33	0.02	0.33	0.26	0.58

Table 3: Per-dataset calibration results.

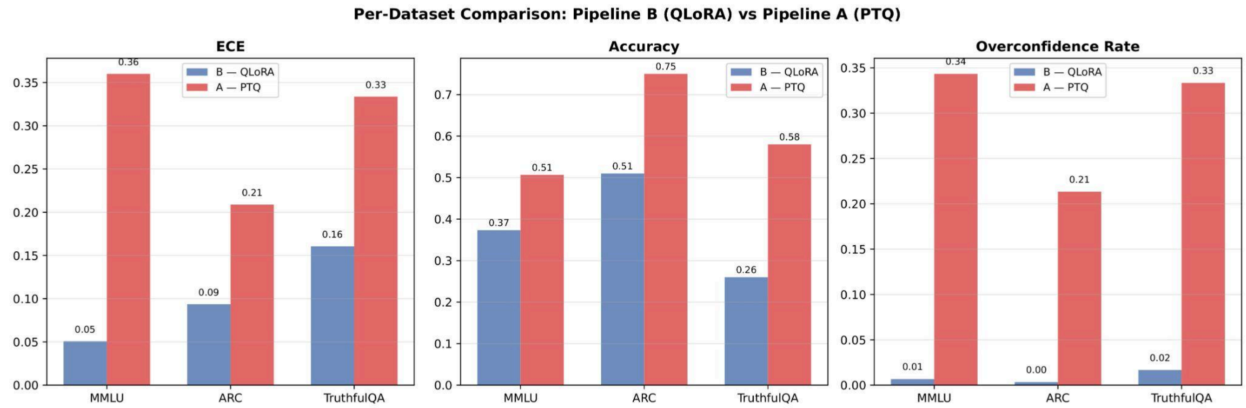

Figure 2: Comparative Performance across MMLU, ARC-Challenge, and TruthfulQA

TruthfulQA shows the most pronounced effect. Pipeline A achieves 58% accuracy on a dataset designed to elicit overconfident false statements, yet exhibits an ECE of 0.33 and a 33% overconfidence rate. This means that for every 3 questions Pipeline A answers incorrectly on TruthfulQA, it expresses high confidence on approximately one of them—precisely the dangerous failure mode the benchmark was designed to detect.

ARC-Challenge is notable for Pipeline B's zero overconfidence rate (0.00). Despite the increased difficulty of multi-step reasoning questions, the QLoRA model never achieves high confidence on a wrong answer. This extreme conservatism is consistent with the undertraining hypothesis discussed in Section 5.

4.3 Calibration-Accuracy Tradeoff

Figure 3 plots the delta accuracy vs. delta ECE for each dataset, where delta is computed as Pipeline A minus Pipeline B. All three datasets fall in the top-right quadrant: Pipeline A gains accuracy at the cost of ECE. The pattern holds without exception, confirming a systematic tradeoff rather than a dataset-specific result.

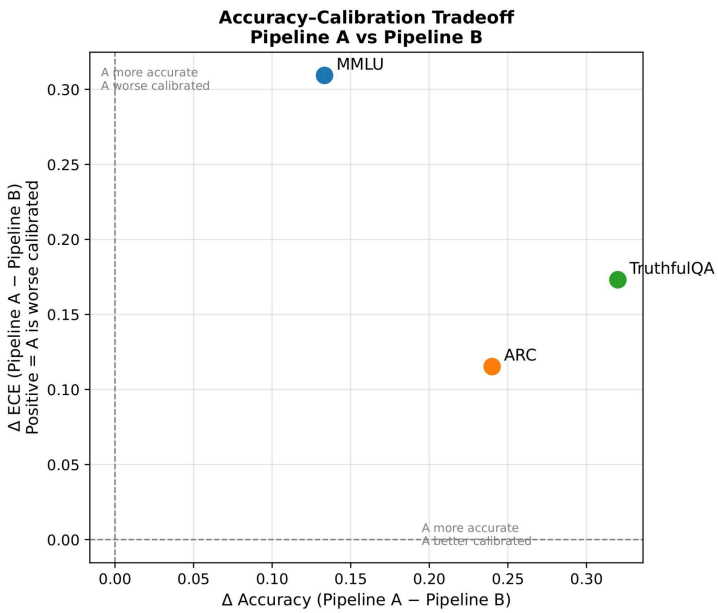

Figure 3: Delta Accuracy vs. Delta ECE for each dataset

This tradeoff has direct implications for deployment decisions. An engineer choosing between pipelines based solely on accuracy metrics would select Pipeline A. Our results show this choice comes with a hidden cost: the selected pipeline is less reliably calibrated about its own uncertainty .

5. Discussion

5.1 The Undertraining-as-Regularization Hypothesis

Why does Pipeline B exhibit dramatically better calibration despite lower accuracy? A plausible explanation is the undertraining-as-regularization effect: a model fine-tuned on 800 samples does not develop the assertive, high-confidence response patterns of a model trained on millions of instruction pairs. The limited training signal is insufficient to push the model toward confident outputs, resulting in a distributional hedging behavior that happens to coincide with good calibration.

This is visible in Figure 4: Pipeline B's confidence distribution is spread across 0.25–0.55 for both correct and incorrect predictions, with mean confidence of 0.412. Pipeline A's distribution is collapsed near 1.0, with the correct and incorrect distributions nearly indistinguishable.

The model has learned to be confident, not to be accurate about when it is confident.

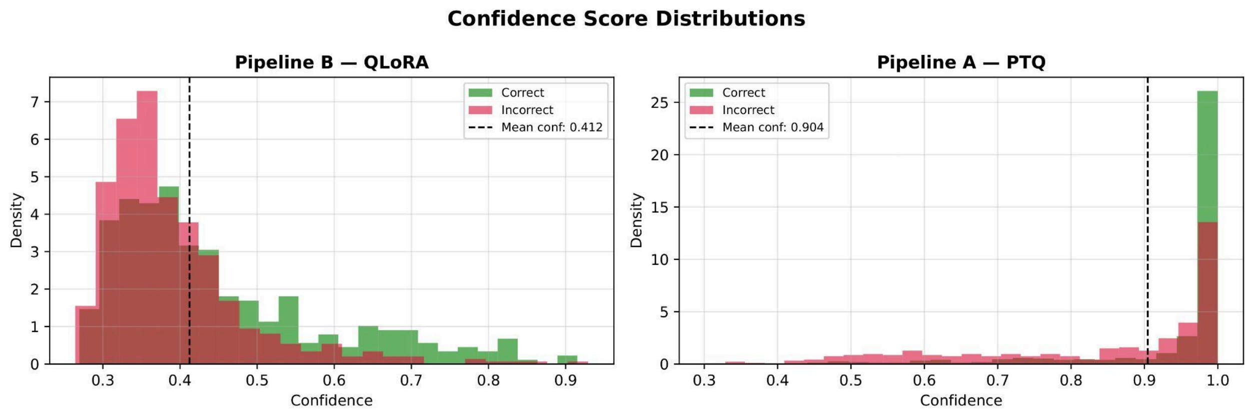

Figure 4: Confidence Score of both pipelines

Importantly, this means Pipeline B's good calibration may be a side effect of undertraining rather than an intrinsic property of the QLoRA quantization order. A QLoRA model trained on the full Alpaca dataset or on professionally curated instruction data might exhibit calibration profiles closer to Pipeline A. We consider this an important open question.

5.2 Safety Implications

The practical implication is direct: the PTQ deployment pipeline, as it is currently practiced—using a professionally instruction-tuned model with post-hoc quantization—is associated with models that exhibit systematically higher overconfidence. In agentic settings where downstream components consume model confidence scores for decision thresholds, routing decisions, or human escalation logic, a model with 29.7% overconfidence rate represents a meaningful safety risk.

TruthfulQA is particularly relevant here. The 33% overconfidence rate on questions specifically designed to expose models' tendency to confidently produce incorrect information means that Pipeline A will, at non-trivial frequency, assert incorrect information with high confidence. This is the worst-case calibration failure for deployed language models.

Pipeline B's primary failure mode—underconfidence (28% rate)—is less dangerous in practice. A model that is correct but uncertain prompts appropriate human review. A model that is wrong but certain does not.

5.3 Implications for Deployment Practice

Our results suggest that the choice between PTQ and QLoRA should incorporate calibration as an explicit criterion alongside accuracy and latency. When deployment context requires reliable uncertainty quantification—medical question answering, legal research assistance, autonomous agents—the calibration gap documented here may outweigh the accuracy advantage of PTQ.

A practical mitigation for PTQ deployment is post-hoc calibration via temperature scaling. Guo et al. (2017) demonstrated that temperature scaling reliably reduces ECE with minimal accuracy impact. Future work should examine whether temperature scaling can close the 7.7x ECE gap observed here.

6. Limitations

We identify three limitations that qualify causal interpretation but do not affect the validity of the empirical comparison.

Instruction tuning confound: The most significant limitation is that pipeline choice is confounded with instruction tuning scale. Pipeline A uses a professionally tuned model (likely trained on millions of instruction pairs); Pipeline B uses 800 Alpaca samples. The calibration difference may reflect instruction tuning data scale rather than quantization order per se. A controlled ablation—fine-tuning the same base model on the same data via PTQ and QLoRA—would isolate quantization order as the independent variable but requires full SFT training infrastructure, which is computationally infeasible on consumer hardware.

Prompt format asymmetry: Pipeline A uses Gemma's chat template; Pipeline B uses the Alpaca instruction format. While each format is appropriate for the respective model's training regime, this introduces a secondary variable. A study using identical prompt formats would provide cleaner comparison.

Sample size: We evaluate on 300 samples per dataset, sufficient for stable ECE estimation with 15 bins but small relative to full benchmark evaluation. Larger evaluation sets may reveal dataset-specific variance not captured here. We use 15 bins rather than the standard 10 to improve bin resolution given our sample size of 300 per dataset.

7. Conclusion

We presented an empirical comparison of confidence calibration in two real-world LLM deployment pipelines: Post-Training Quantization (PTQ) and QLoRA-based fine-tuning. Using Gemma-2-2B as the base model and evaluating across MMLU, ARC-Challenge, and TruthfulQA, we found that the evaluated PTQ pipeline configuration exhibits a 7.7x higher Expected Calibration Error and a 33x higher overconfidence rate than the QLoRA pipeline, despite achieving 23% higher task accuracy.

The central finding is simple: accuracy and calibration are not jointly optimized by current deployment practices. The pipeline that yields higher accuracy also exhibits substantially higher overconfidence. For production systems where calibration safety matters—agents, medical applications, high-stakes decision support—this tradeoff is non-trivial and currently invisible to standard evaluation practice.

We propose that calibration metrics, particularly overconfidence rate, should become standard components of LLM deployment evaluation alongside accuracy and latency. The 4-bit quantized models that dominate edge deployment deserve the same calibration scrutiny as their full-precision counterparts.

Ignoring calibration in deployment is not a neutral choice, it implicitly selects for overconfident behavior.

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